

Template 1: Direct Proof

Goal: $\forall x(P(x) \rightarrow Q(x))$. **Structure:** Let x be arbitrary. Assume $P(x)$, show $Q(x)$.

1. State the Theorem:

2. Identification:

Hypothesis $P(x)$:

Conclusion $Q(x)$:

3. The Proof:

1. *The Setup*

"Let x be an arbitrary element of . Assume that..."

(Write $P(x)$ here)

2. *Unpacking Definitions*

Translate $P(x)$ to equations, other propositions. ("Put things on the table.")

3. *The creative part*

Manipulate equations/objects on the table to reach the form of $Q(x)$.

4. *Repacking Definitions*

Conclude that you satisfied the definition of $Q(x)$.

5. *Conclusion*

"Since x was arbitrary, we have shown that $\forall x(P(x) \rightarrow Q(x))$."

□

Template 2: Proof by Contrapositive

Goal: $\forall x(P(x) \rightarrow Q(x))$. **Structure:** Let x be arbitrary. Assume $\neg Q(x)$, show $\neg P(x)$.

1. **State the Theorem:**

2. **Identification:**

Negate Conclusion ($\neg Q(x)$):

Negate Hypothesis ($\neg P(x)$):

3. **The Proof. “We show the contrapositive.”**

1. *The Setup*

”Let x be an arbitrary element of . Assume that...”

(Write $\neg Q(x)$ here)

2. *Unpacking Definitions*

Translate $\neg Q(x)$ into equations, other propositions. (“Put things on the table.”)

3. *The creative part*

Manipulate equations/objects on the table to reach $\neg P(x)$.

4. *Repacking Definitions*

Show matches definition of $\neg P(x)$.

5. *Conclusion*

”Since x was arbitrary, we conclude $\forall x(P(x) \rightarrow Q(x))$, as desired.”

□

Template 3: Proof by Contradiction

Goal: $\forall x(P(x) \rightarrow Q(x))$. **Structure:** Let x be arbitrary. Assume $P(x) \wedge \neg Q(x)$, conclude F .

1. **State the Theorem:**

2. **Identification:**

Assumption ($P(x)$):

Negate Conclusion ($\neg Q(x)$):

3. **The Proof. “We proceed by contradiction.”**

1. *The Setup (The Trap)*

”Suppose for contradiction there exists $x \in$ such that...”

(Write “ $P(x) \wedge \neg Q(x)$.”)

2. *Unpacking Definitions*

Translate $P(x) \wedge \neg Q(x)$ into equations, other propositions. (“Put things on the table.”)

3. *The creative part*

Show they lead to a contradiction.

4. *The Contradiction*

State the contradiction clearly.

5. *Conclusion*

”Thus, $\forall x(P(x) \rightarrow Q(x))$.”

□